In leaves

- Leaves are generally leathery and shiny to reflect light and heat.
- In some plants like *Euphorbia, Acacia, Ziziphus* and *Capparis*, the stipules are modified into spines.
- The entire leaves are modified into spines (*Opuntia*) or reduced to scales (*Asparagus*).

Anatomical adaptations

- Presence of multilayered epidermis with heavy cuticle to prevent water loss due to transpiration.
- Hypodermis is well developed with sclerenchymatous tissues.
- Sunken stomata are present only in the lower epidermis with hairs in the sunken pits.
- Scotoactive type of stomata found in succulent plants .
- Vascular bundles are well developed with several layered bundle sheath.
- Mesophyll is well differentiated into palisade and spongy parenchyma.
- In succulents the stem possesses a water storage region.

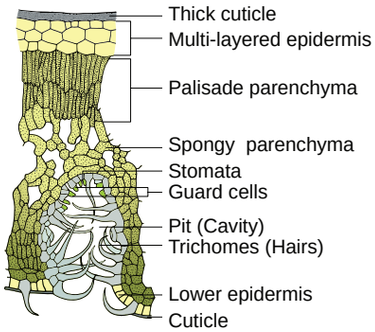


Figure 6.23: T.S. of Nerium leaf

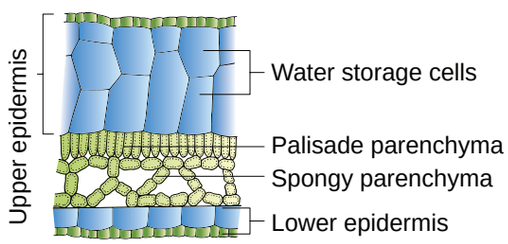


Figure 6.24: A Succulent leaf of *Peperomia* (T.S.) (lateral wing portion only)

Physiological adaptations

- Most of the physiological processes are designed to reduce transpiration.
- Life cycle is completed within a short period (Ephemerals).

Mesophytes

The plants which are living in moderate conditions (neither too wet nor too dry) are known as **mesophytes**. These are common land plants. Example: Maize and *Hibiscus.*

Morphological adaptations

- Root system is well developed with root caps and root hairs
- Stems are generally aerial, stout and highly branched.
- Leaves are generally large, broad, thin with different shapes.

Anatomical adaptations

- Cuticle in aerial parts are moderately developed.
- Epidermis is well developed and stomata are generally present on both the epidermis.
- Mesophyll is well differentiated into palisade and spongy parenchyma.
- Vascular and mechanical tissues are fairly developed and well differentiated.

Physiological adaptations

- All physiological processes are normal.
- Temporary wilting takes place at room temperature when there is water scarcity.

Tropophytes are plants which behave as xerophytes at summer and behave as mesophytes (or) hydrophytes during rainy season.

Epiphytes

Epiphytes are plants which grow perched on other plants (Supporting plants). They use the supporting plants only as shelter and not for water or food supply. These epiphytes

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